S&S QUIZ-4 SOLUTION

Sol 1 (a):- Given –

$$y(t) = \int_{-\infty}^{t} e^{-(t-\tau)} \cdot x(\tau-3)\, d\tau$$

Let $\tau - 3 = \lambda$

$d\tau = d\lambda$

$$y(t) = \int_{-\infty}^{t-3} e^{-(t-\lambda-3)} \cdot x(\lambda)\, d\lambda \quad \rightarrow (1 \text{ Mark})$$

$$= \int_{-\infty}^{\infty} e^{-(t-\lambda-3)} \cdot u(t-\lambda-3) \cdot x(\lambda)\, d\lambda \quad \text{—— (1)}$$

$\rightarrow (1 \text{ Mark})$

As we know that – $y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\lambda$ —— (2)

$\rightarrow (1 \text{ Mark})$

Comparing eqn (1) & eqn (2), we have –

Impulse response, $h(t) = e^{-(t-3)} \cdot u(t-3)$ $\rightarrow (2 \text{ Mark})$

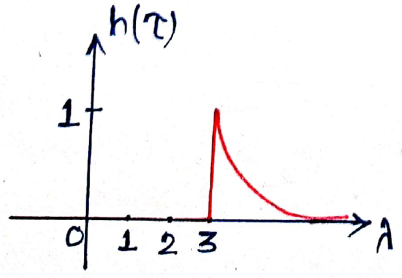
Sol 1 (b):- Given – $x(t) = u(t+1) - u(t-2)$

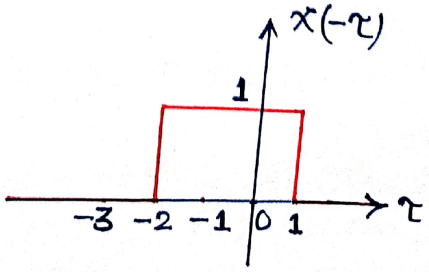
$h(t) = e^{-(t-3)} \cdot u(t-3)$

Impulse Response of the system, $y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\lambda$

$$= \int_{-\infty}^{\infty} x(t-\tau) \cdot h(\tau)\, d\lambda$$

$\rightarrow (1 \text{ Mark})$





$\rightarrow (2 \times 0.5 \text{ Mark})$

Scanned with OKEN Scanner

Case(1):- For $t \leq 2$, $y(t) = 0$ → (1 Mark)

Case(2):- For $2 < t \leq 5$, $y(t) = \int_3^{t+1} e^{-(\tau-3)} d\tau$

$$= \left\{ \frac{e^{-(\tau-3)}}{-1} \right\}_3^{t+1}$$

$$= -\left\{ e^{-(t+1-3)} - e^{-(3-3)} \right\}$$

$$= 1 - e^{-(t-2)}$$ → (1 Mark)

Case(3):- For $5 < t$, $y(t) = \int_{t-2}^{t+1} e^{-(\tau-3)} d\tau$

$$= \left\{ \frac{e^{-(\tau-3)}}{-1} \right\}_{t-2}^{t+1}$$

$$= -\left\{ e^{-(t+1-3)} - e^{-(t-2-3)} \right\}$$

$$= e^{-(t-5)} - e^{-(t-2)}$$ → (1 Mark)

Scanned with OKEN Scanner